Practice workbook · Build or read a pedigree to decide how a condition is inherited

Drawing a family tree from a description, and using pedigree patterns to judge dominant versus recessive and autosomal versus sex-linked inheritance.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INFER-INHERITANCE-MODE-FROM-PEDIGREE

1. Question 8 · part b.i · 4 marks

June 2019 Paper 2R

Worked-example retry · Attempt part b.i only.

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8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

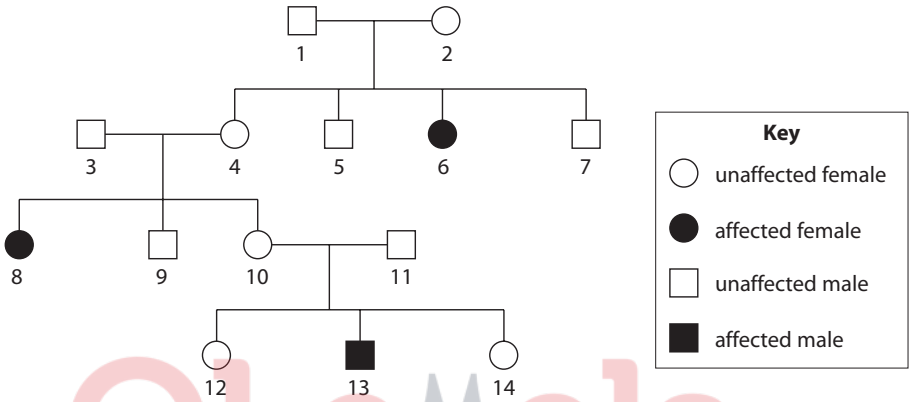
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1. Question 8 · part b.i · 4 marks

June 2019 Paper 2R

Worked-example retry · Attempt part b.i only.

(b) The diagram shows the pedigree of a family affected by alkaptonuria.



(i) Use the family pedigree to explain whether the alkaptonuria condition is sex-linked.

(4)

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1. Question 8 · part b.i · 4 marks

June 2019 Paper 2R

Worked-example retry · Attempt part b.i only.

(ii) Parents 10 and 11 have another child.

Calculate the probability that this child would be an affected male.

Explain your answer.

(3)

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(Total for Question 8 = 13 marks)

1. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

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2. Question 8 · part b.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

8 (a) Genetic conditions are often caused by recessive alleles.

(i) Which of these describes an allele?

(1)

- ☒ **A** a chromosome that controls a characteristic
- ☒ **B** an alternative form of a gene
- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

.....

.....

(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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P 6 7 0 6 2 R A 0 2 6 2 8

2. Question 8 · part b.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(ii) Discuss the conclusions the student can make about the inheritance of the condition.

(3)

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(Total for Question 8 = 8 marks)

Working and answer

3. Question 7 · part a.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

7 Huntington's chorea is a dominant genetic condition that results in the nervous system not working correctly. It is not a sex-linked condition.

Diagram 1 shows a family pedigree. Some members of the family have the condition.

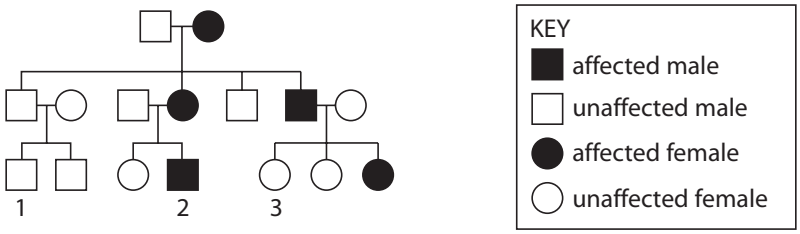


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 7 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

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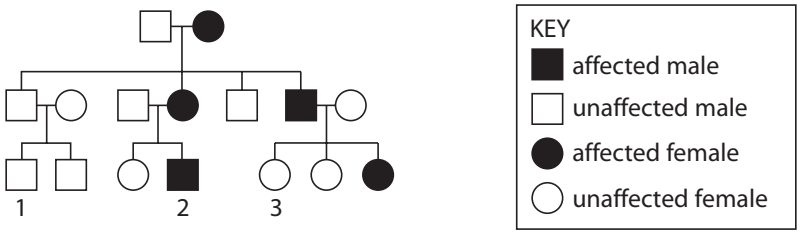


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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4. Question 7 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

(b) (i) A man has the same genotype as person 1 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

These two people are expecting a child.

Calculate the probability that the child would be a boy not showing the Huntington's condition, giving reasons for your answer.

(3)

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4. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 8 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

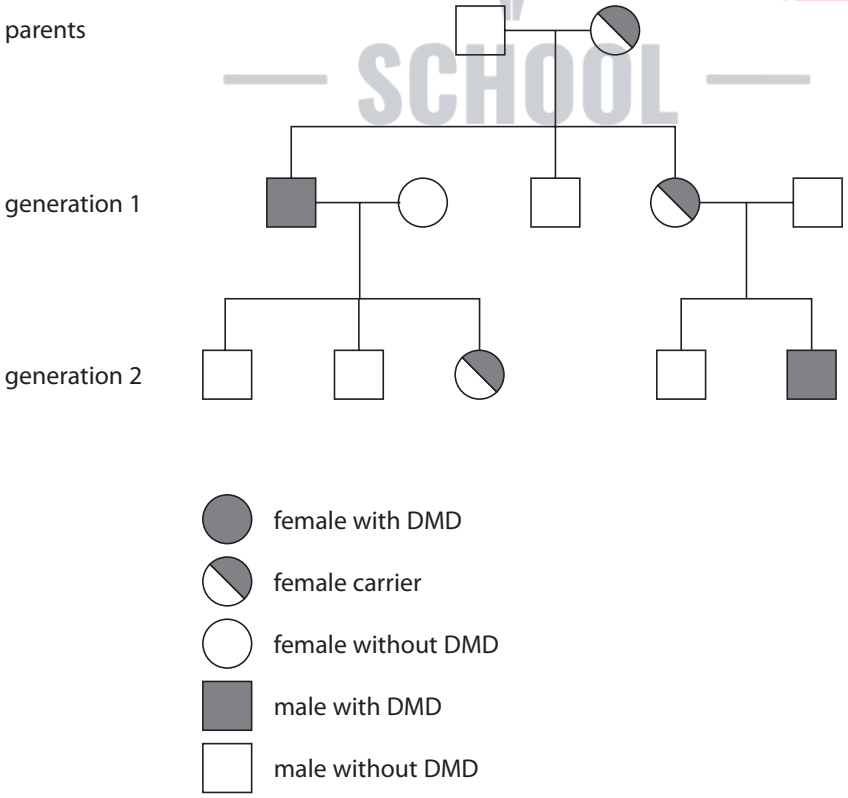
8 (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

(4)

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.



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5. Question 8 · part b.ii · 3 marks

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Further practice · Attempt part b.ii only.

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(i) Explain what is meant by a carrier of DMD.

(2)

(ii) Explain how the diagram shows that DMD is caused by a recessive allele.

(3)

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(iii) Draw a genetic diagram to show the inheritance of DMD from the parents to generation 1.

Use X^D for the dominant allele and X^d for the recessive allele.

(2)

(Total for Question 8 = 11 marks)

5. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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6. Question 3 · part b.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.ii only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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6. Question 3 · part b.ii · 1 mark

June 2024 Paper 1

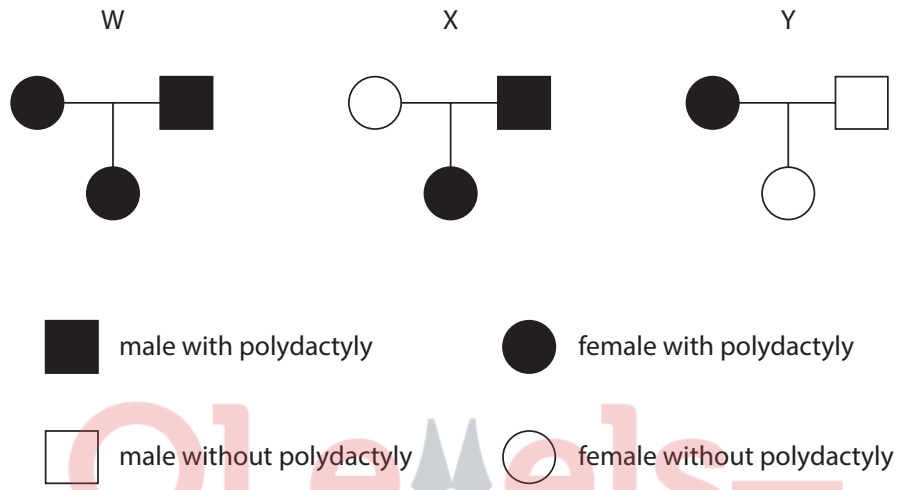
Further practice · Attempt part b.ii only.

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(ii) The family pedigree diagrams show the inheritance of polydactyly.



Which of the diagrams, W, X or Y, gives the family pedigree for parents R and S?

(1)

(c) A gene is a short section of DNA.

Describe the structure of DNA.

(4)

(Total for Question 3 = 9 marks)

6. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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7. Question 5 · part b.ii · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

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(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

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P 7 5 9 5 8 A 0 1 2 2 4

7. Question 5 · part b.ii · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

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(b) (i) Haemolytic anaemia (HA) is an inherited condition.
People with this condition have a low number of red blood cells in their blood.

Explain one symptom that people with HA will have.

(4)

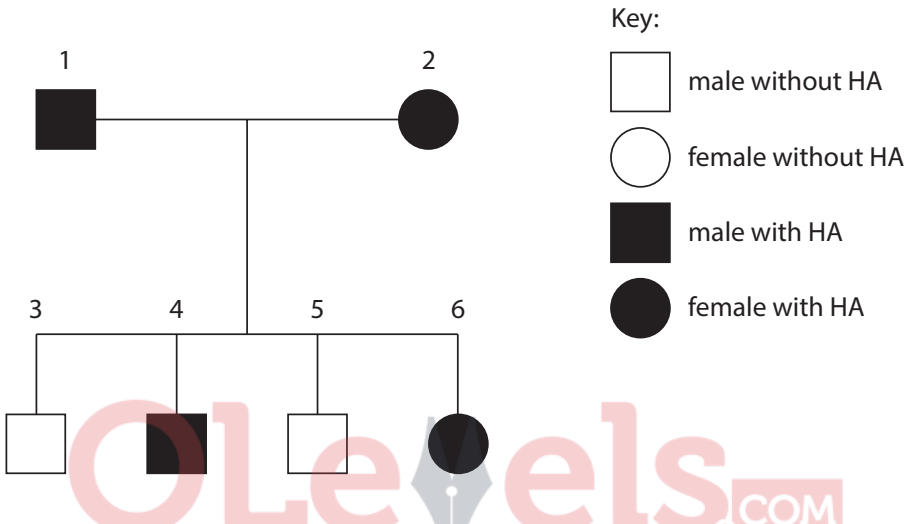
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7. Question 5 · part b.ii · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

(ii) The diagram shows the inheritance of HA in one family.



Explain how HA is inherited.

(4)

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(Total for Question 5 = 14 marks)

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7. Question 5 · part b.ii · 4 marks · Working and answer

Working and answer

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8. Question 8 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

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(i) Which of these describes an allele?

(1)

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- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

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(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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8. Question 8 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

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(ii) Discuss the conclusions the student can make about the inheritance of the condition.

(3)

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(Total for Question 8 = 8 marks)

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Working and answer